

Lab 4 - Vibe Coding Conversation

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Software Engineering | Assigned game: SETAPESU26/08_helicopter

AI assistant: Codex | 7 October 2026

Edited and consolidated from the actual project discussion. The opening brief expands the original request using the supplied handout; later prompts paraphrase real feedback. This is not a verbatim transcript.

1. Project brief and approach

Student - expanded project brief

Please complete Software Engineering Lab 4, Vibe Coding, using the supplied handout and the game repository assigned to my SRN, PES1UG24AM920. Confirm the correct assignment before changing the starter. The goal is to repair and complete the existing small Pygame game, while preserving its basic structure.

Use the assigned README as the task specification. Fix the movement and boundary bug: prolonged key presses must not create unlimited vertical speed, direction changes should respond promptly, and the complete player body must remain inside the top and bottom boundaries. Implement obstacle collision with a clear game-over state, distance scoring with reset on a new game, and a visibly active shield that absorbs one obstacle collision.

Run the unchanged starter and record ten seconds before editing. Complete the tasks individually, with separate commits, then record ten seconds demonstrating the fixed movement and added features. Verify the running game and videos rather than relying only on code inspection. Put the updated code, videos and conversation document in Lab-4 of my personal repository; do not raise a pull request to the instructor repository. Maintain a local state file so the project can be resumed. Accepted Labs 1-3 must remain unchanged.

Assistant

I checked the assignment and identified SETAPESU26/08_helicopter as your starter. Its scope is one movement/boundary bug and three features: collision, distance scoring and a shield. I will follow those requirements, record the original game before making changes, and keep the work separate from the accepted lab submissions.

The implementation order will be movement, collision/game over, scoring, and shield behavior. I will verify each task, preserve separate task commits, and keep progress and resume information in the working state file.

2. Implementation progress

Assistant - movement and boundaries

The original recording shows the movement problems: speed builds while a direction key is held, reversal is delayed, and the player can disappear below the screen. I capped vertical speed, discarded velocity in the old direction when the opposite key is pressed, and added release damping. The boundary checks now contain the whole player rectangle at both edges. The four initial movement checks pass.

Assistant - collision and scoring

Wall contact now triggers a game-over state, while clear passage through the gap remains safe. Updates stop after game over, and R starts a fresh game. Distance increases with the scrolling movement, the final distance is displayed when the game ends, and restart resets the score.

Assistant - initial shield and recording

Space activates the shield, and a blue outline indicates protection. The initial version consumes the shield on the first wall contact and uses contact tracking to avoid treating the same overlap as a second hit. This initial contact model will be revised after the video review below.

All four tasks have separate commits. The initial behavioral suite has 14 passing tests. The before and after recordings have been prepared, and I am checking that the videos demonstrate the required behavior and match the submitted source.

3. Deliverable and video review

Student

Please check that every required deliverable is present, especially the gameplay videos. I have not seen the recordings in the handoff, so confirm that the before and after videos were made and show where I can inspect them.

Assistant

Both videos were recorded and uploaded, but I had not displayed them in the earlier reply. I checked their playback, confirmed that each is exactly ten seconds, and verified that the local files match the repository copies. I then showed both recordings for review.

The package contains the updated game, individual task commits, before/after videos, and a conversation document. The before video retains the unchanged starter so the original movement problem can be compared with the completed version.

4. Correction based on gameplay feedback

Student

The after video still looks incorrect. The player appears to move down out of view and pass through solid walls. Please review the video and the running game as many times as necessary. Confirm the boundaries and collision behavior in actual gameplay, and update the demonstration only after the result is clear and reliable.

Assistant

The review identified a problem with the shield contact model. After protection was consumed, continued overlap with the same solid wall was still treated as an absorbed contact. This made the player appear to pass through a wall that remained visibly intact. I changed the shield-hit response to clear the struck wall and consume protection immediately; subsequent solid-wall contact remains lethal.

I also moved the HUD below the playfield so it would not obscure the player near the bottom boundary. The checks were expanded to 18 cases, including exact edge contact, safe gap positions, long-held movement keys, shield consumption, restart and score reset. Revised native recordings were reviewed for the boundary, shield-hit and unprotected-collision transitions.

5. Simplification and manual testing

Student

Keep the solution proportionate to this simple game. Revisit the assigned deliverables, fix the actual issues, and remove unnecessary code or complicated behavior. I would also like to play the original and completed versions so I can compare the controls and collisions directly.

Assistant

I simplified the shield response rather than retaining the additional per-wall state and hit timer. In the final version, an absorbed hit consumes the shield and removes the hit obstacle. The next obstacle is lethal unless the player activates protection again. This removes 33 runtime lines from the preceding revision, and the obstacle class returns to the original starter implementation.

All 18 behavioral tests pass. I recorded a matching after video showing bottom containment, upward reversal, a safe unshielded gap crossing, one shield hit, a later unprotected collision, and restart with score reset.

Both native versions were launched for manual comparison. Up/Down controls movement in each. In the completed game, Space activates the shield and R restarts after game over. The windows were labelled BEFORE and AFTER to make the comparison clear.

6. Assignment confirmation and final scope

Student

Please confirm that this is the exact game assigned to me. I cannot see a helicopter image, and I am unsure why a shield is included. Explain the original appearance, whether the shield is required, and the expected final deliverables. Keep all submission files consistent with those requirements.

Assistant

The official Section H assignment sheet maps PES1UG24AM920 to game number 8 and SETAPESU26/08_helicopter. The original renderer draws the player as a dark rounded rectangle; no helicopter image is supplied. Retaining that appearance is consistent with the starter.

The shield is explicitly required by Task 4. It must be visibly active, protect against one obstacle collision, and turn off immediately after use. Space is the activation control in this implementation. Removing the hit obstacle is an implementation choice for resolving the protected contact, not a separate instructor requirement.

The handout requires updated code, a ten-second before video, a ten-second after video, and chat history as a DOC/PDF under Lab-4 in the personal repository. The starter additionally asks for a conversation-page link. Each task has its own commit, and no pull request is made to SETAPESU26.

7. Submission presentation

Student

Please keep the repository easy for the teacher to evaluate. Avoid multiple Markdown documents, keep working notes outside the submission, and present the project discussion in a clear, professional format.

Assistant

The teacher-facing package now uses one README, the updated game source, both videos, and this edited conversation PDF. The README links to the PDF preview. Working state, original records and intermediate checks are kept outside Lab-4.